MD. NUR ALAM AKASH

+88015-21515249 | nurakash16@gmail.com

LinkedIn | Portfolio | GitHub



CAREER SUMMARY

Software Engineer with hands-on experience building industrial IoT dashboards, monitoring platforms, and maintenance management systems using Angular, TypeScript, Node.js, and PostgreSQL. Experienced in the full cycle from user research and UI/UX design to frontend development, API integration, and database design. Skilled in real-time telemetry platforms (ThingsBoard, MQTT, Node-RED) and in collaborating with cross-functional teams on enterprise client projects. Electrical and Electronics Engineering graduate with a background in VLSI design, which provides a strong analytical and debugging foundation.

WORK EXPERIENCE

Software Engineer, Sincos Group — Feb 2025 – Present

Dashboards and Frontend

- Develop industrial monitoring and reporting dashboards using Angular, TypeScript, JavaScript, HTML, and Tailwind CSS.
- Contributed to dashboards for CMMP (Compressor Monitoring), Energy Meter, Flow Meter, Carbon Emission, and Generator Efficiency systems.
- Contributed to the UI/UX upgrade of the Production Evaluating System (PES) dashboard frontend, redesigning layouts and fixing monitoring issues in collaboration with the team.
- Improved UI responsiveness, layout consistency, and navigation flow across the CMMP, EMS, and FMS platforms.
- Designed multi-screen enterprise dashboards for Unilever with a focus on clear data visualization and usability.
- Worked on dashboards for the LADWP project.
- Developed dashboards for the Gaugap agritech project and worked with the data path from MQTT to ThingsBoard.

IoT Integration and Backend

- Developed the full Angular Gateway application for communication and control of IoT and PLC devices.
- Worked on MQTT-to-ThingsBoard data flows, using Node-RED and REST APIs to ingest, retrieve, and visualize real-time telemetry.
- Worked with ThingsBoard assets, entities, devices, roles, and permissions, and called its APIs from custom dashboards.
- Collaborated with iCenter modules to integrate data and enhance dashboard interactions.

Maintenance Management System

- Contributed to the Maintenance Management System across UI design, Angular frontend, API integration, and database design.
- Researched competitor CMMS platforms (ClickMaint) and documented findings to guide feature and UX improvements.

Other Contributions

- Built a custom Markdown editor with rich-text formatting.
- Added multilingual support (Bangla and French) for client deployment flexibility.
- Support operations at Savar Continental by assessing workflows and PES usage, improving dashboards and software, and conducting user training.

Chip Designer (Trainee), Nokshasemi — Nov 2023 – Nov 2024

- Completed an industrial training program in CMOS circuit design, schematic development, and layout using Cadence Virtuoso and Assura.
- Designed and verified 65nm CMOS circuits (ring oscillator, frequency divider, counter, flip-flop, op-amp), achieving DRC-clean and LVS-clean layouts through systematic debugging and simulation.
- Applied timing analysis, parasitic analysis, and signal integrity checks to validate designs against rule-based specifications.

SKILLS

- **Frontend Development:** Angular, TypeScript, JavaScript, HTML5, CSS3, Tailwind CSS, React (basic), WebRTC (basic)
- **Backend and Databases:** Node.js, REST API design and integration, PostgreSQL, Firebase, database design
- **IoT and Industrial Integration:** ThingsBoard (telemetry, assets, devices, roles, permissions), MQTT, Node-RED, IoT/PLC gateways
- **UI/UX and Dashboards:** Figma, industrial dashboard layouts, data visualization, user-flow planning
- **Tools and Deployment:** Git/GitHub, Docker Compose (basic), Nginx (basic), Linux CLI, LLM API (basic)
- **Programming and Engineering:** Python, C/C++, MATLAB, Verilog HDL, CMOS/VLSI design (Cadence Virtuoso/Assura), DRC/LVS verification
- **Core Strengths:** Problem-solving, communication, fast learning, teamwork, adaptability

PROJECTS

- **Real-Time Chat Application:** Chat app with a React web version, built on Firebase and WebRTC. Supports text, emoji, voice messages, image, audio and document sharing, and voice and video calls.
- **Finance Tracker:** Full-stack finance app (Angular, Node.js, PostgreSQL) with data visualization and an AI assistant that answers questions about spending. Runs with Docker Compose.
- **IELTS Practice Platform:** Free full-length mock exam website that simulates the real test format.
- **Graduate Records System:** Full-stack application for managing and displaying graduate information.
- **ISP Movie Streaming Website:** Streaming platform built for a local internet service provider.
- **VLSI Projects:** 16-bit MIPS processor (placement and routing), 4-bit ripple carry adder, low-power flip-flop.
- **Embedded Projects:** Arduino-based solar traffic light, blind assistance system and AC fan speed controller; wireless DC motor control.

ACHIEVEMENTS

- Runners-up, VLSI Hackathon, MindSparks 2022 (powered by Neural Semiconductor)
- Top 15 finalists, VLSI Day 2022 Design Competition, UIU
- Final Year Design Project Poster Exhibition (FYDP): selected in the top 10%

RESEARCH

- **Research Article:** A high-performance biosensor design for waterborne bacteria detection based on one-dimensional photonic crystal
- **Thesis:** Optical properties of a defective one-dimensional photonic crystal containing graphene nanolayers

EDUCATION

B.Sc. in Electrical & Electronics Engineering — **Mar 2018 – Nov 2022**
Ahsanullah University of Science & Technology | CGPA 3.24

Higher Secondary Certificate, Science — **Jun 2015 – Apr 2017**
Dhaka City College | CGPA 5.00

EXTRACURRICULAR ACTIVITIES

- General member, AUST Cultural Club (helped organize events with the executive board)
- Organizer, AUST EEE Day Spring 2021; Co-Organizer, AUST EEE Excursion 2019
- Helped organize the Annual Office Tour at Sincos Group

REFERENCES

Md. Faysal Nayan, Assistant Professor, Dept. of EEE,
Ahsanullah University of Science & Technology
faysal.nayan.eee@aust.edu

Aman Ullah, Head of Industrial IoT, Sincos Automation Technologies Ltd.
aman@sincosbd.com